

Ethanol Blending in Gasoline – Luxembourg

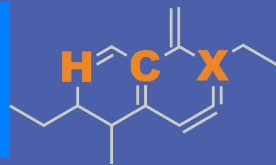
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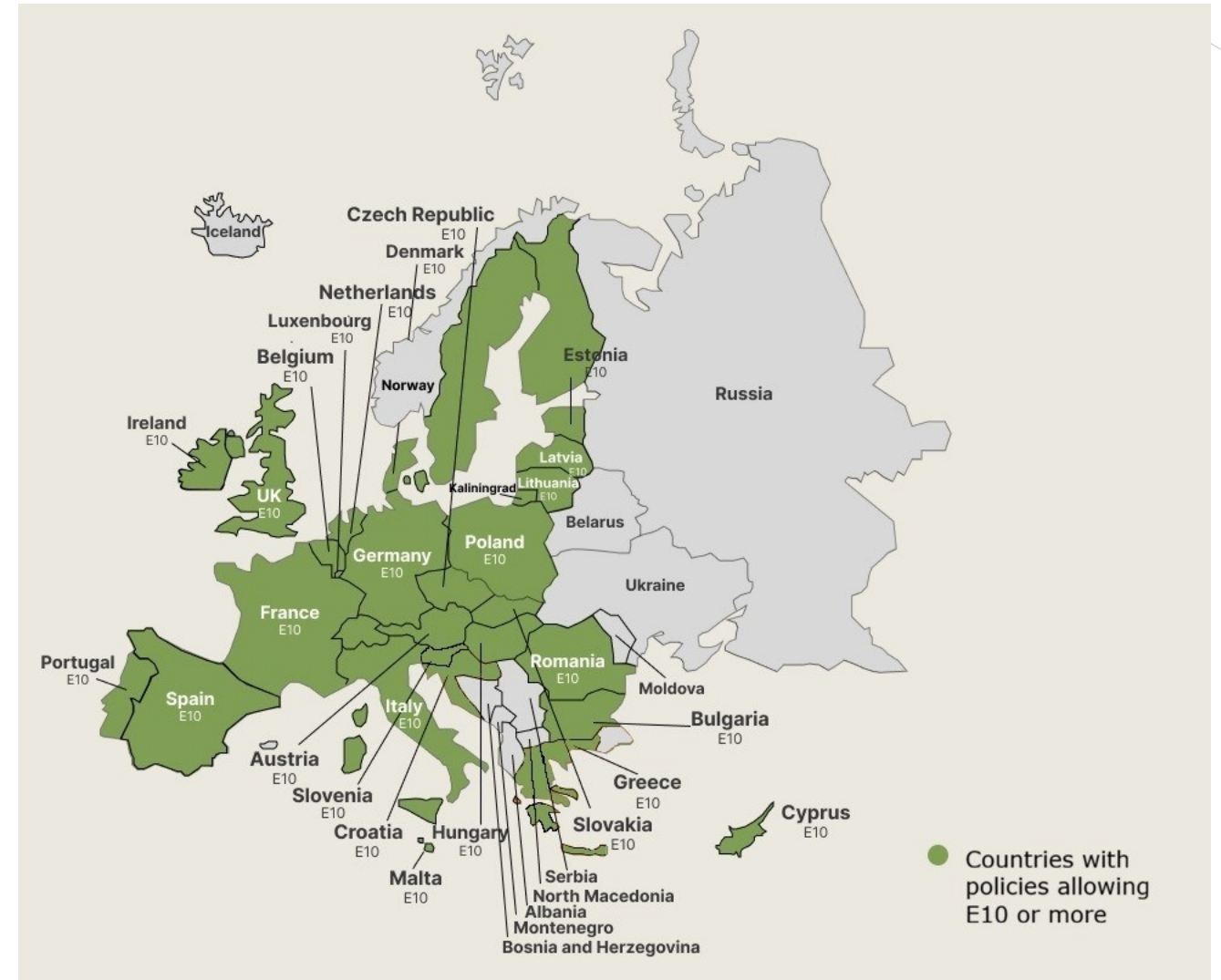


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Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

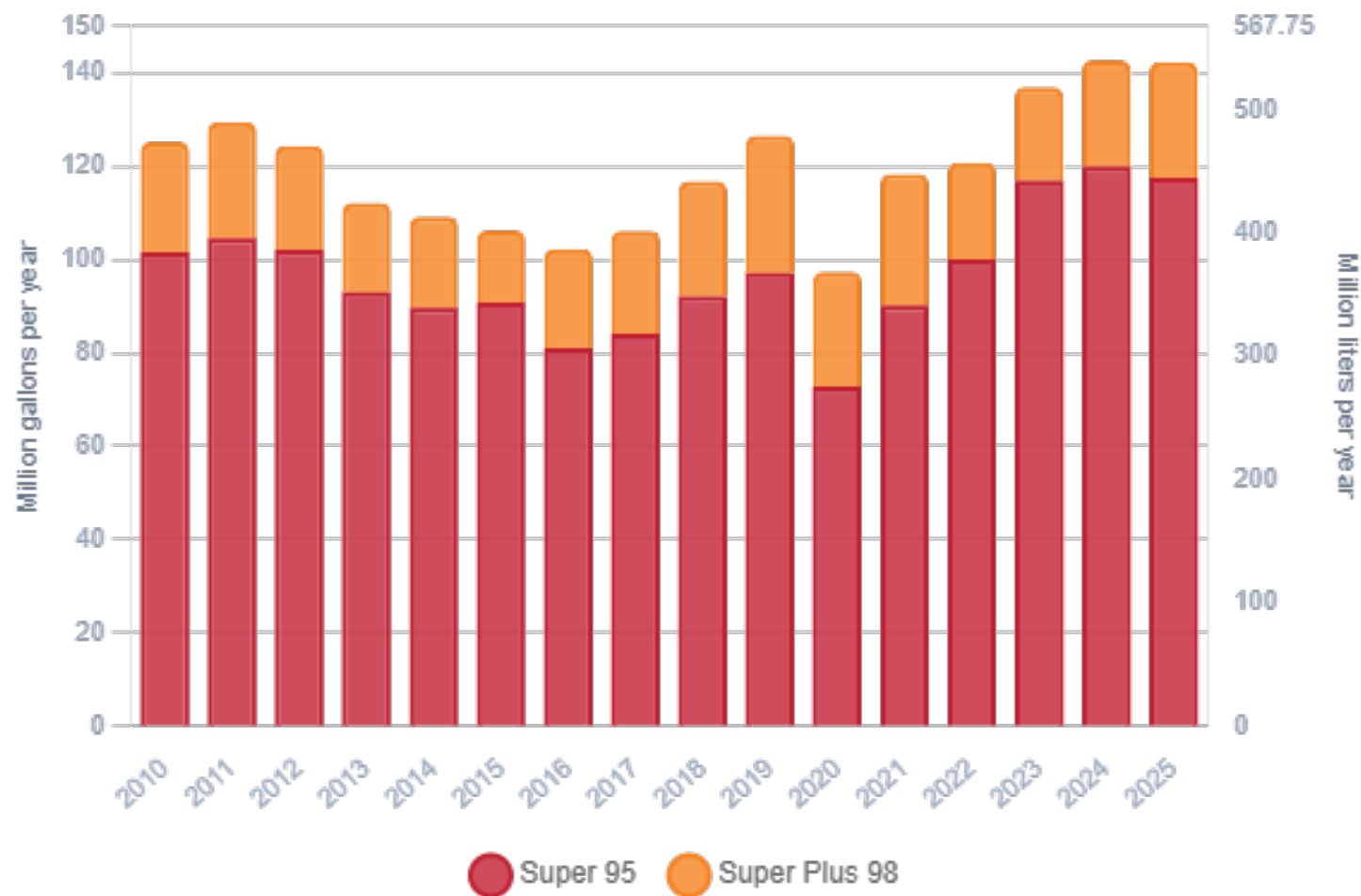
Ethanol Gasoline Blending in Luxembourg



In 2025, gasoline consumption in Luxembourg was 539 million liters (142 million gallons) and growth rate was 0.86%. There is no gasoline production. Gasoline net imports were 485 million liters (128 million gallons) correspondingly. Luxembourg complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.6 RON.

In 2025, ethanol consumption in Luxembourg was 24 million liters (6 million gallons) representing 4% of gasoline demand. Net Imports were 27 million liters (7 million gallons). Ethanol sources are Sugar Cane 80% and Corn 20%.

Gasoline Demand in Luxembourg



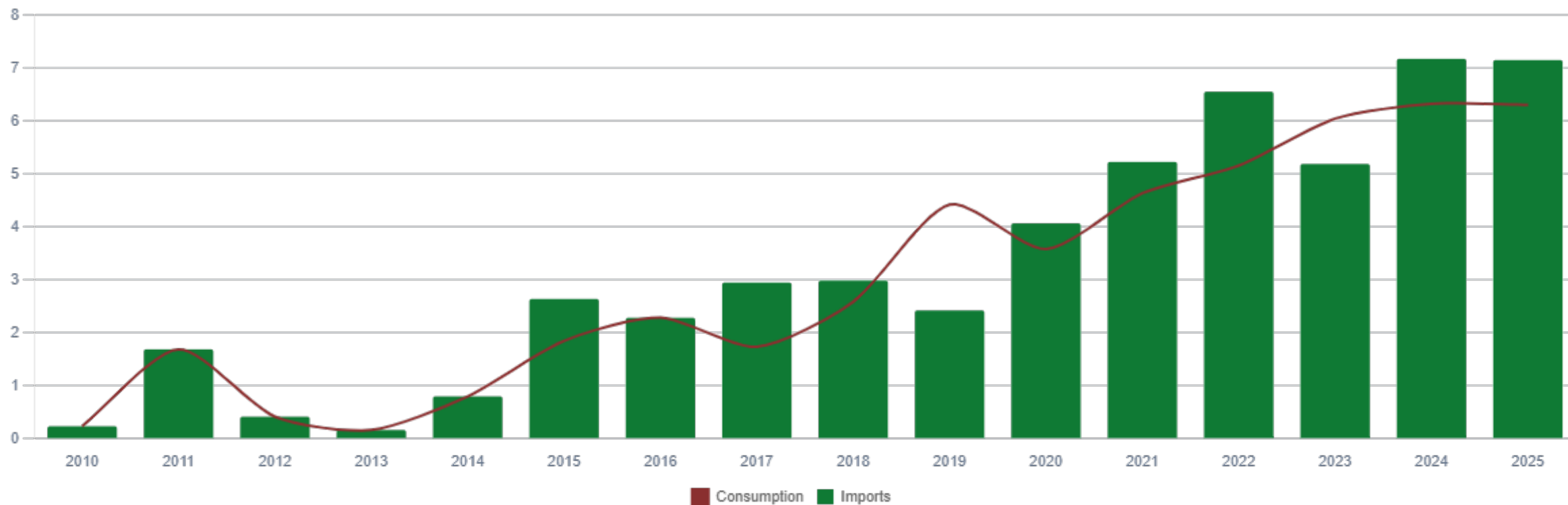
Source: Eurostat

Gasoline Imports in Luxembourg



Source: Eurostat

Ethanol Balance in Luxembourg



Ethanol is supplied only by imports

Source: Eurostat

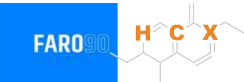
Gasoline Quality in Luxembourg

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Countries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	7.7							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	6.0							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commission

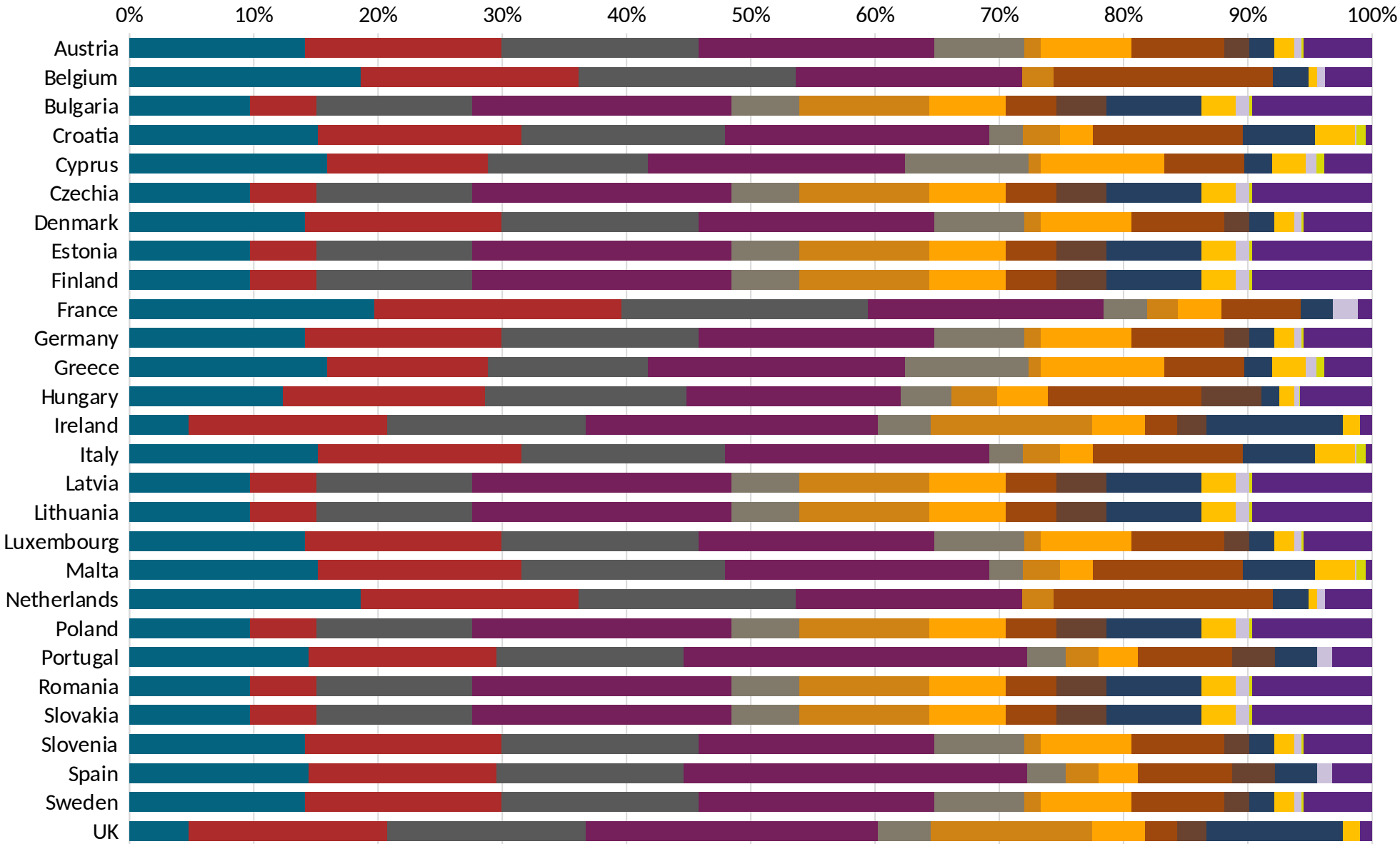
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source:
Faro90



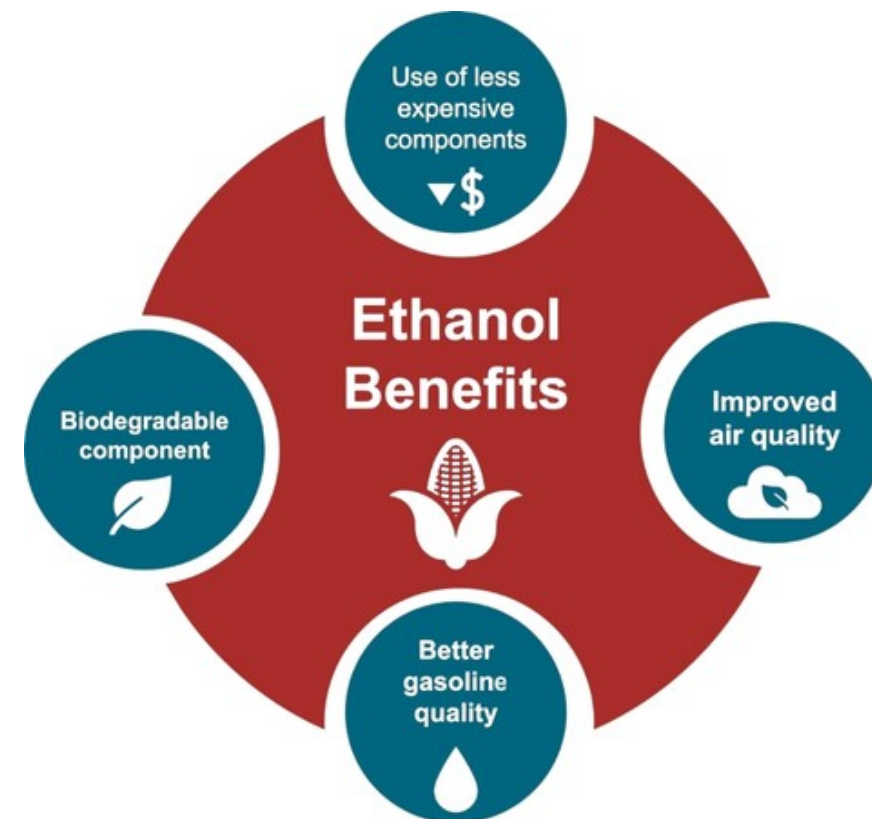
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

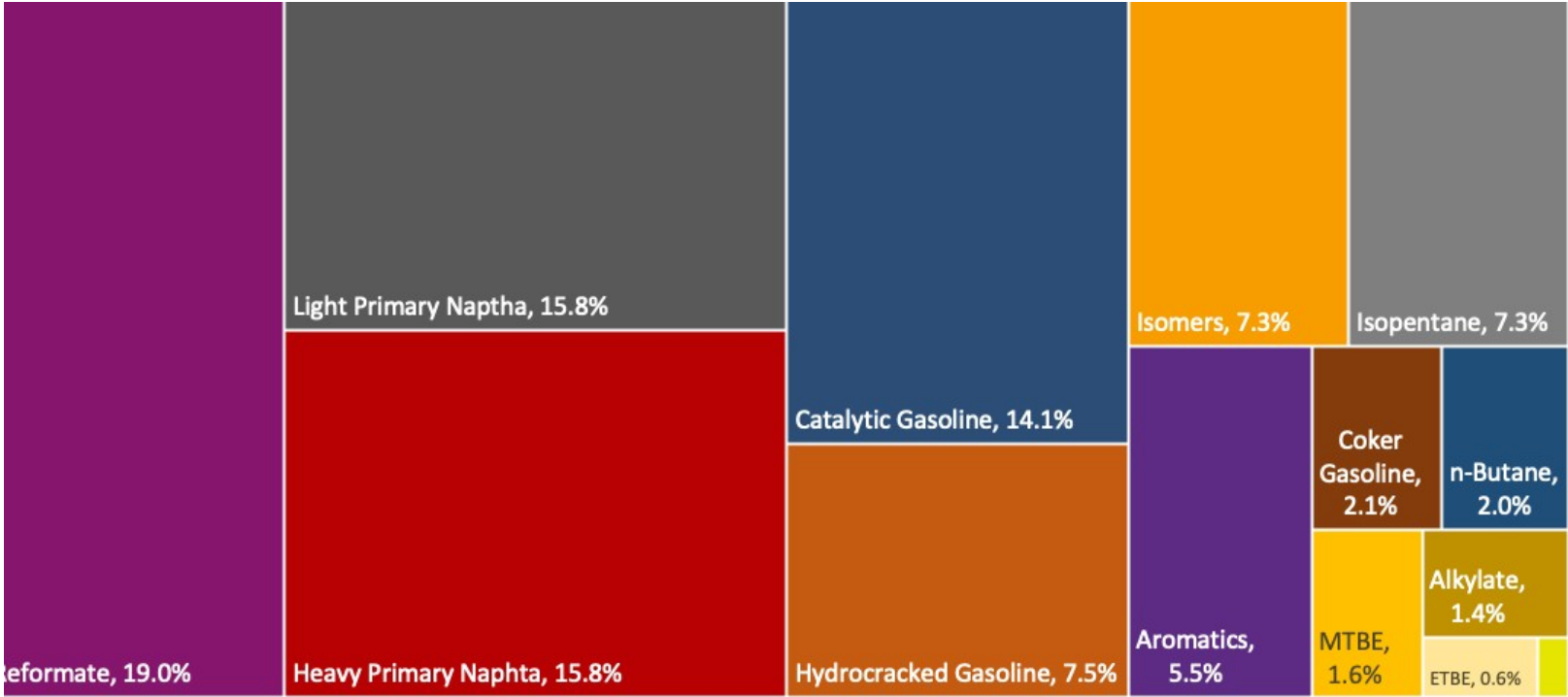
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

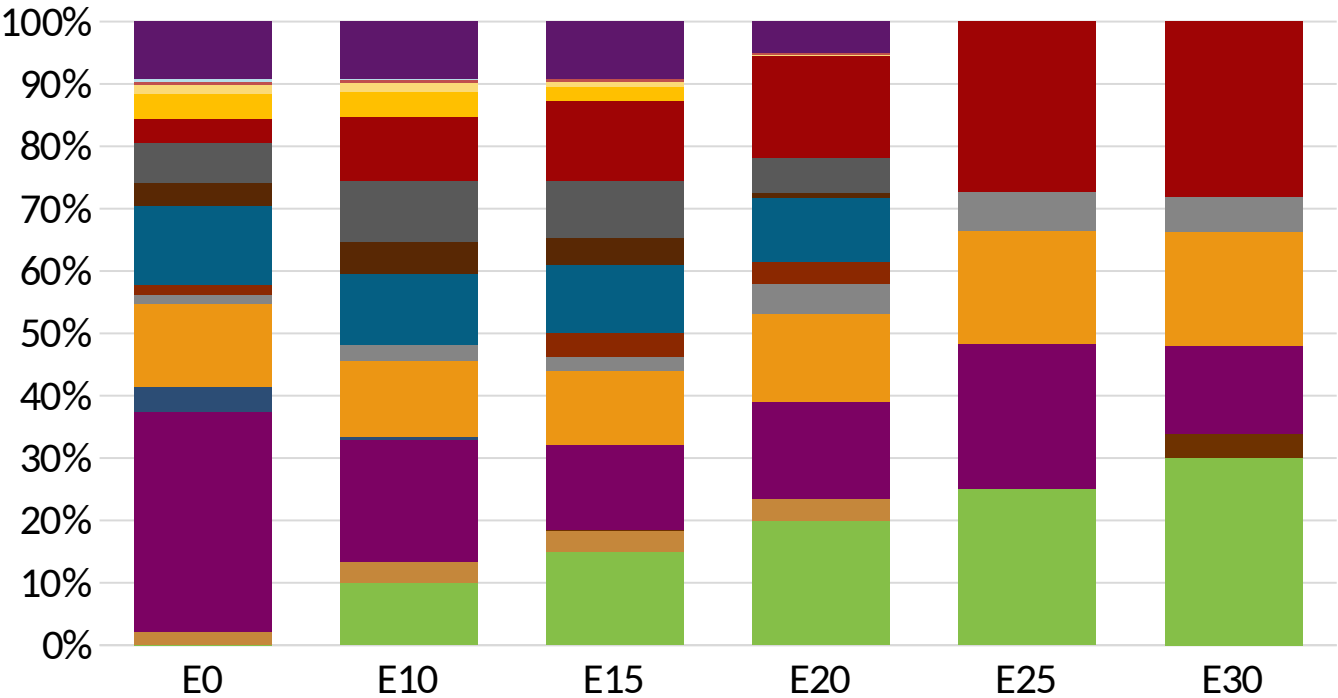
The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Luxembourg Available Blending Components



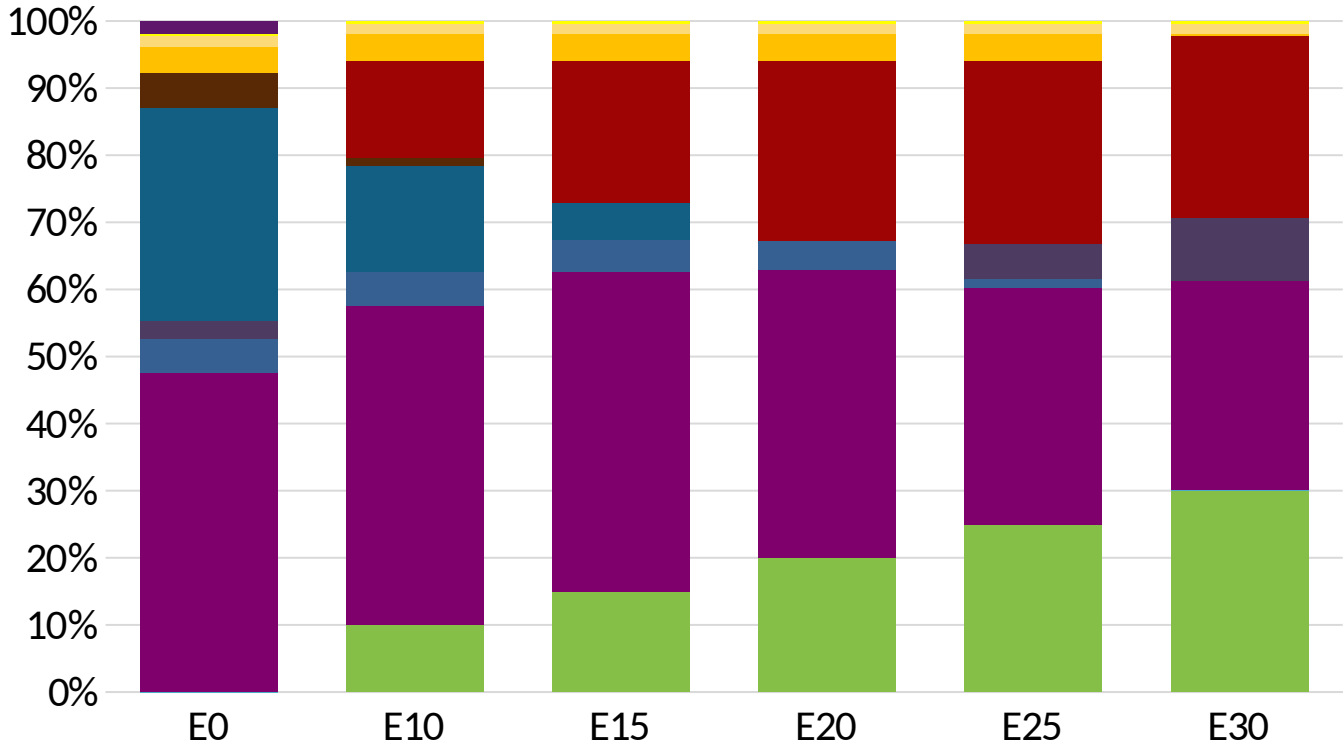
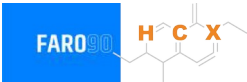
Luxembourg – Gasoline 95 – Constant Octane



Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US\$/gal)	\$ 2.069	\$ 1.941	\$ 1.880	\$ 1.799	\$ 1.677	\$ 1.612

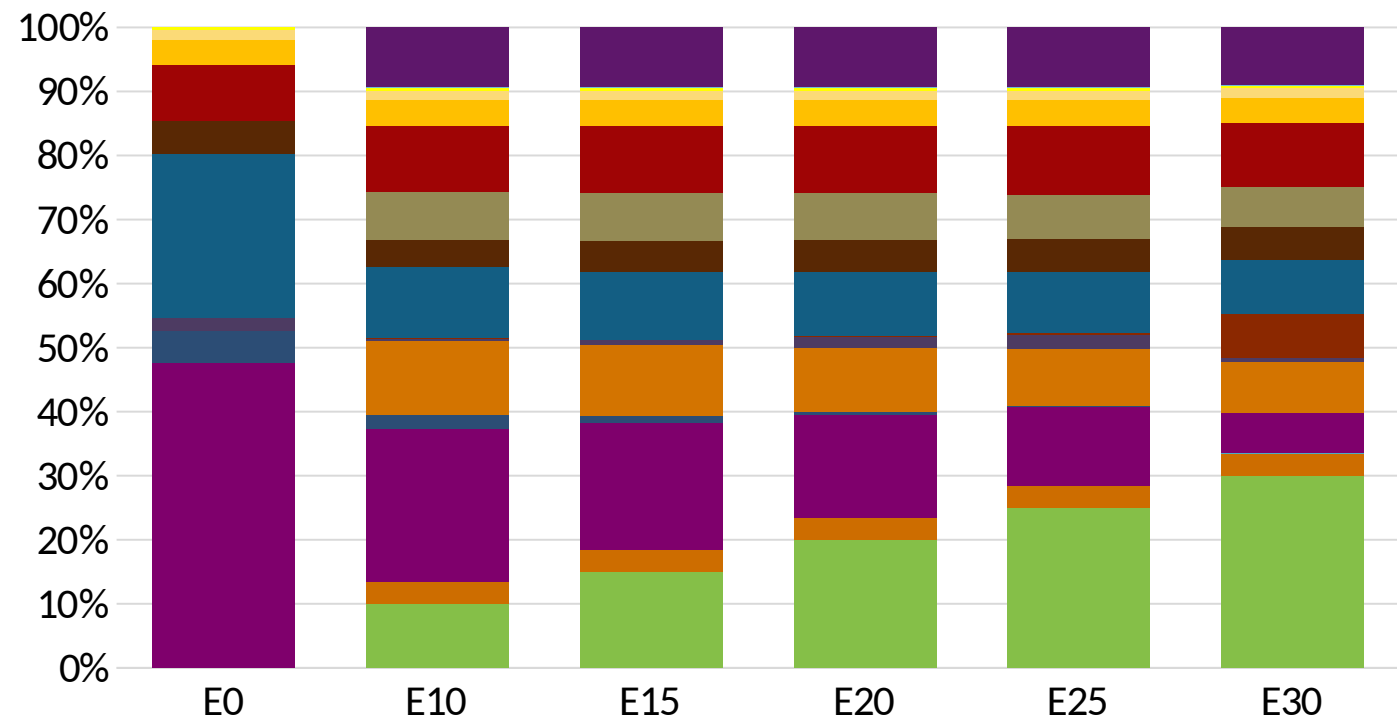
Luxembourg - Gasoline 98 - Constant Octane



Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.144	\$ 2.015	\$ 1.943	\$ 1.866	\$ 1.789	\$ 1.722

Average CIF 2025 Prices

Luxembourg – Gasoline 95 – Increased Octane

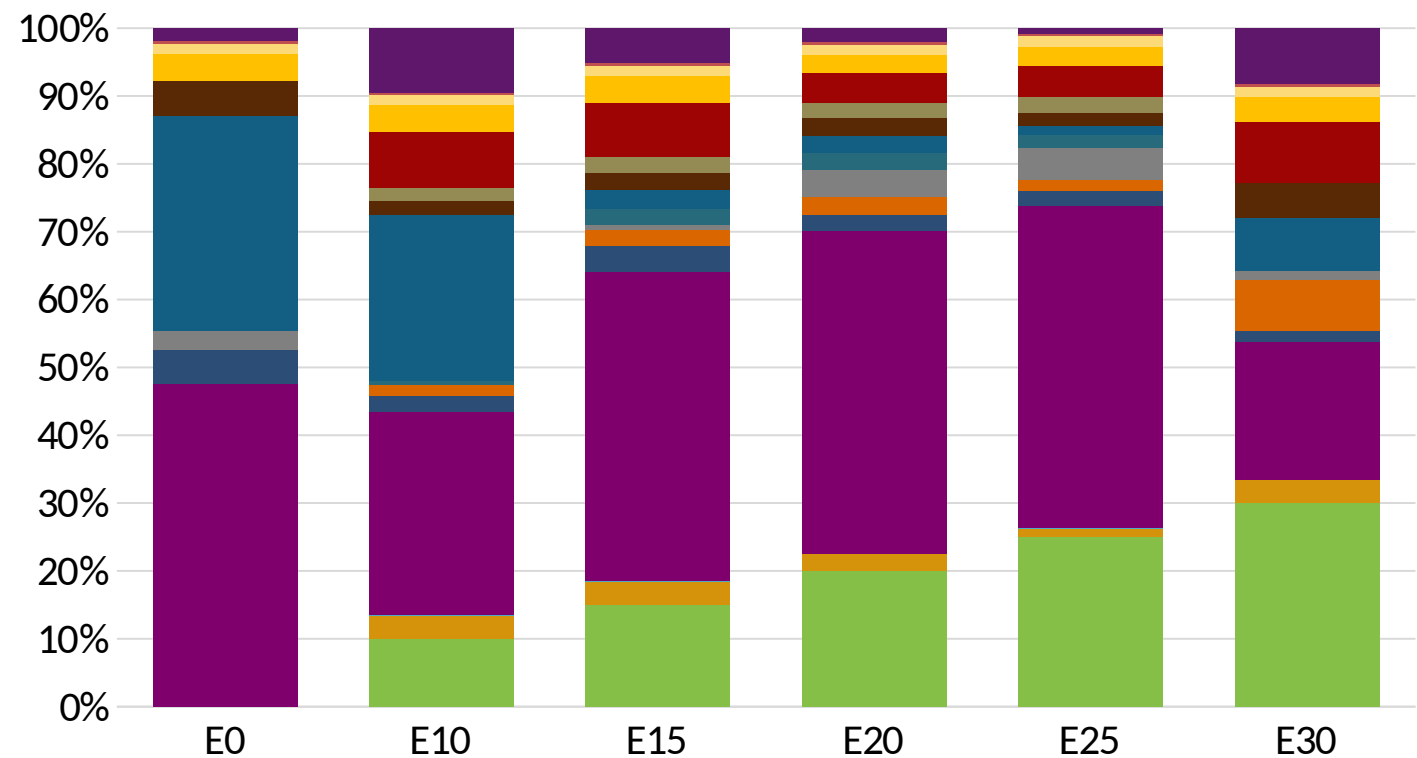


Average CIF 2025
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US\$/gal)	\$ 2.006	\$ 2.006	\$ 2.006	\$ 2.006	\$ 2.006	\$ 2.006

Source:
Faro90

Luxembourg – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	98.0	99.2	101.3	102.6	104.0	104.7
Price (US\$/gal)	\$ 2.144	\$ 2.144	\$ 2.144	\$ 2.144	\$ 2.144	\$ 2.144

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

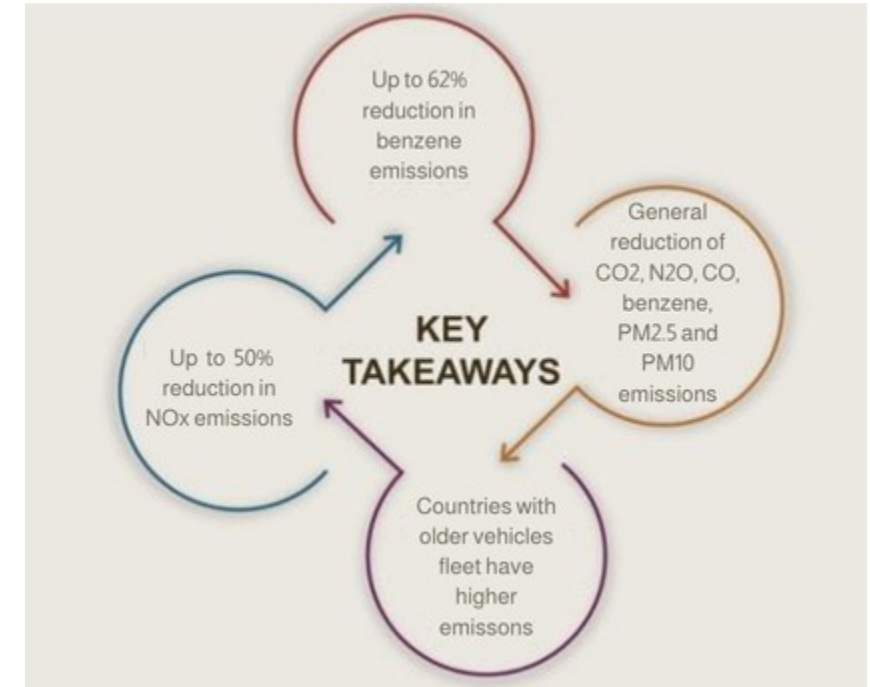
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

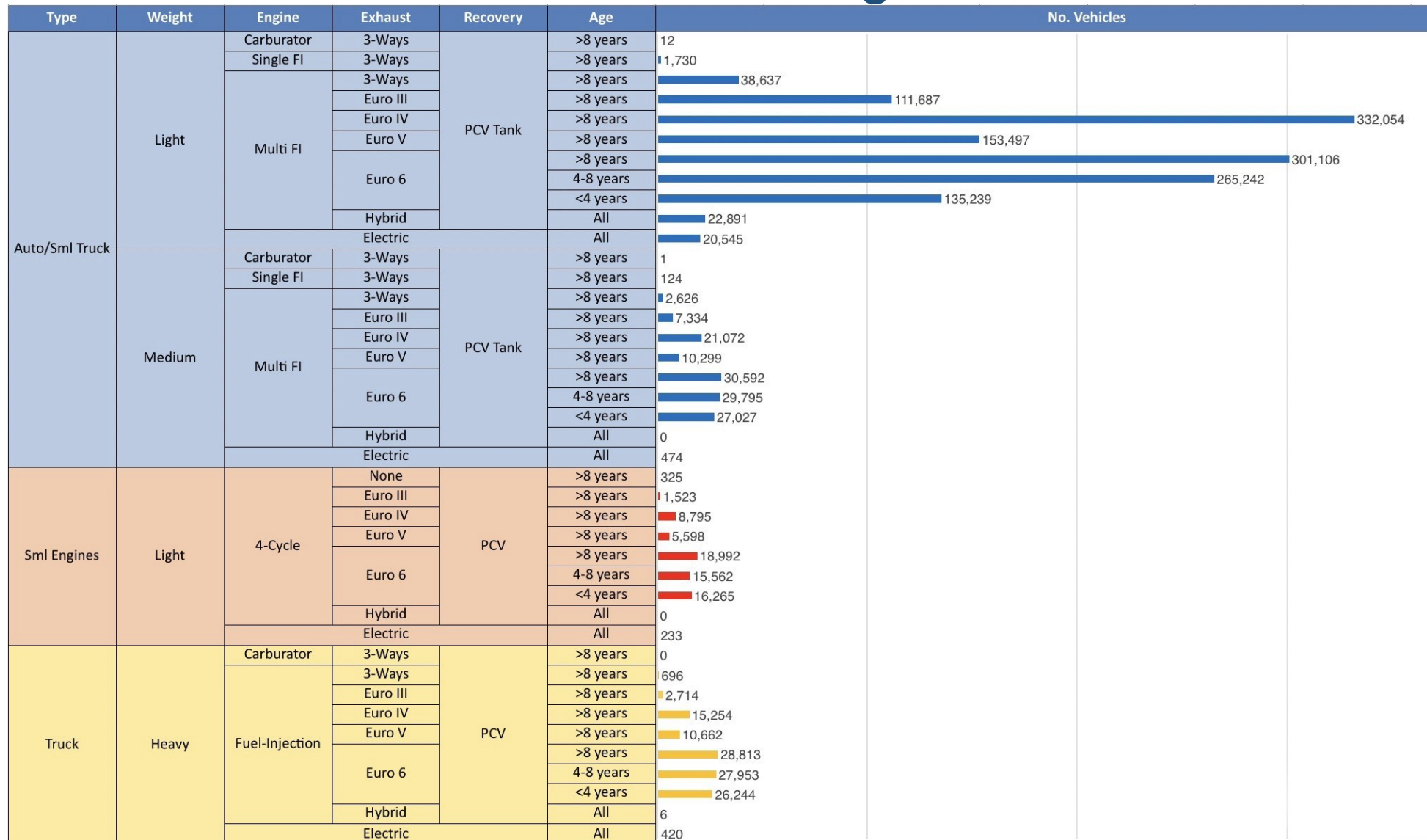
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Luxembourg

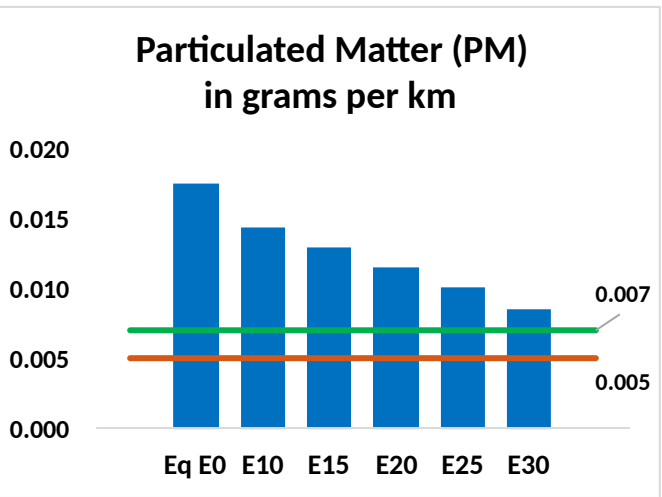
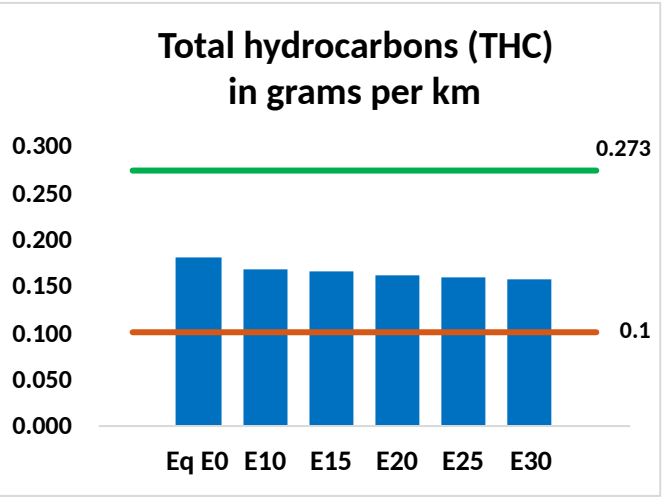
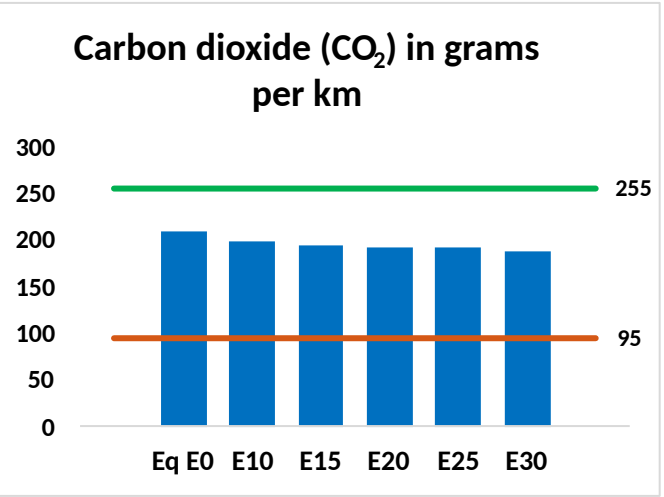
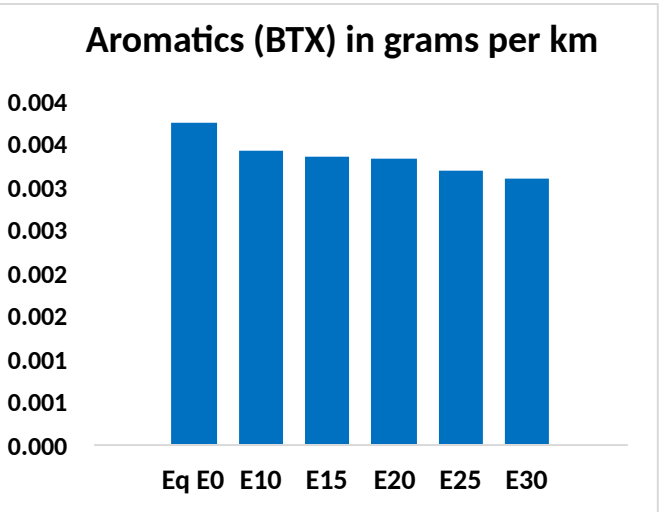
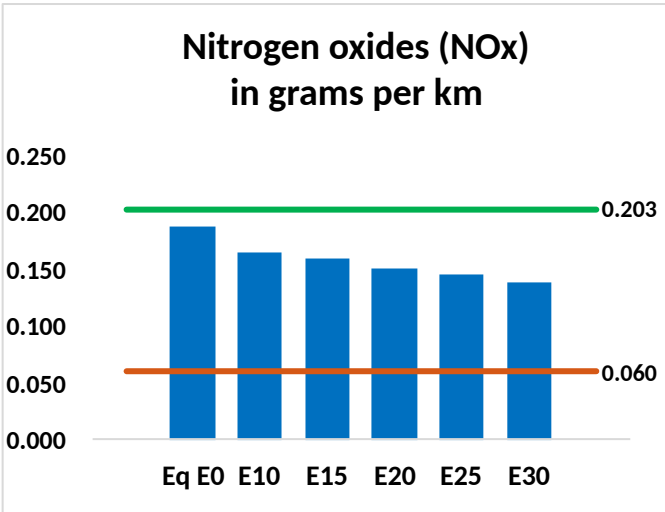
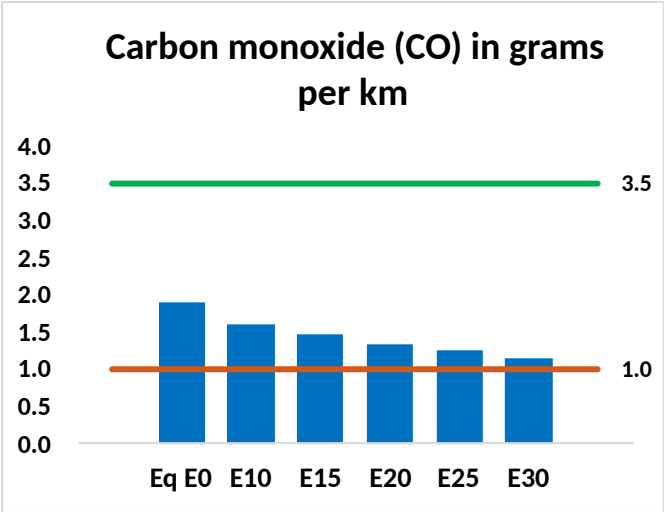
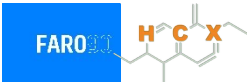


Vehicle Fleet: 1,692,040
Gasoline Vehicle Fleet: 1,150,362

Source: Eurostat, ACEA

Luxembourg – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Luxembourg – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	58,155	53,465	48,776	44,890	40,960	37,933	34,697	-16%	-30%
CO2	6,391,079	6,226,737	6,071,525	5,949,455	5,889,050	5,852,719	5,744,836	-5%	-8%
VOC	5,518	5,330	5,142	5,033	4,946	4,885	4,781	-7%	-10%
NOx	5,779	5,336	5,037	4,887	4,642	4,452	4,227	-13%	-20%
SOx	73	67	62	57	53	48	43	-15%	-28%
NH3	2,172	2,150	2,129	2,139	2,163	2,180	2,194	-2%	0%
N2O	73	72	72	72	74	75	76	-1%	2%
CH4	1,324	1,324	1,324	1,344	1,373	1,398	1,422	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	63	61	60	59	58	58	57	-6%	-8%
Acetaldehyde	120	137	201	307	418	477	564	68%	249%
Formaldehyde	349	339	395	464	486	538	585	13%	39%
Benzene	115	110	105	103	102	98	95	-9%	-11%
PM2.5	382	348	313	282	252	220	187	-18%	-34%
PM10	153	139	125	113	101	88	75	-18%	-34%

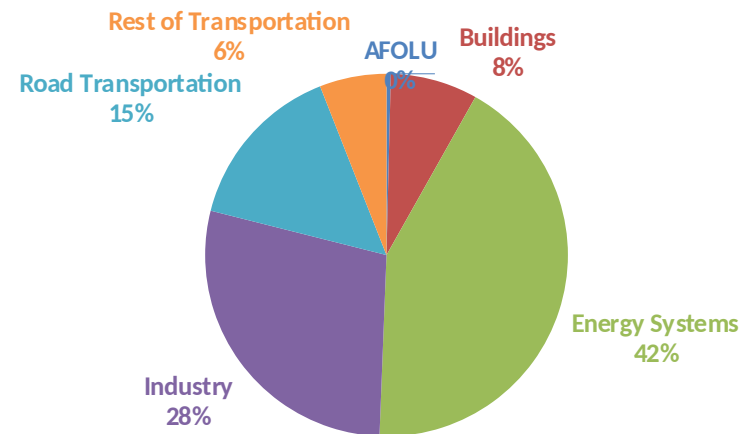
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

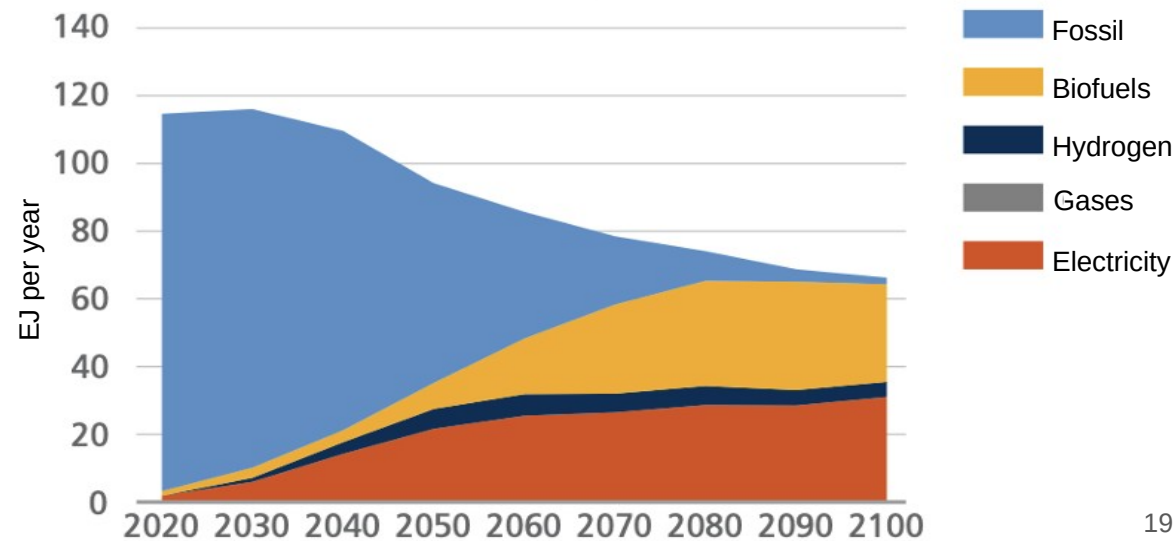
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

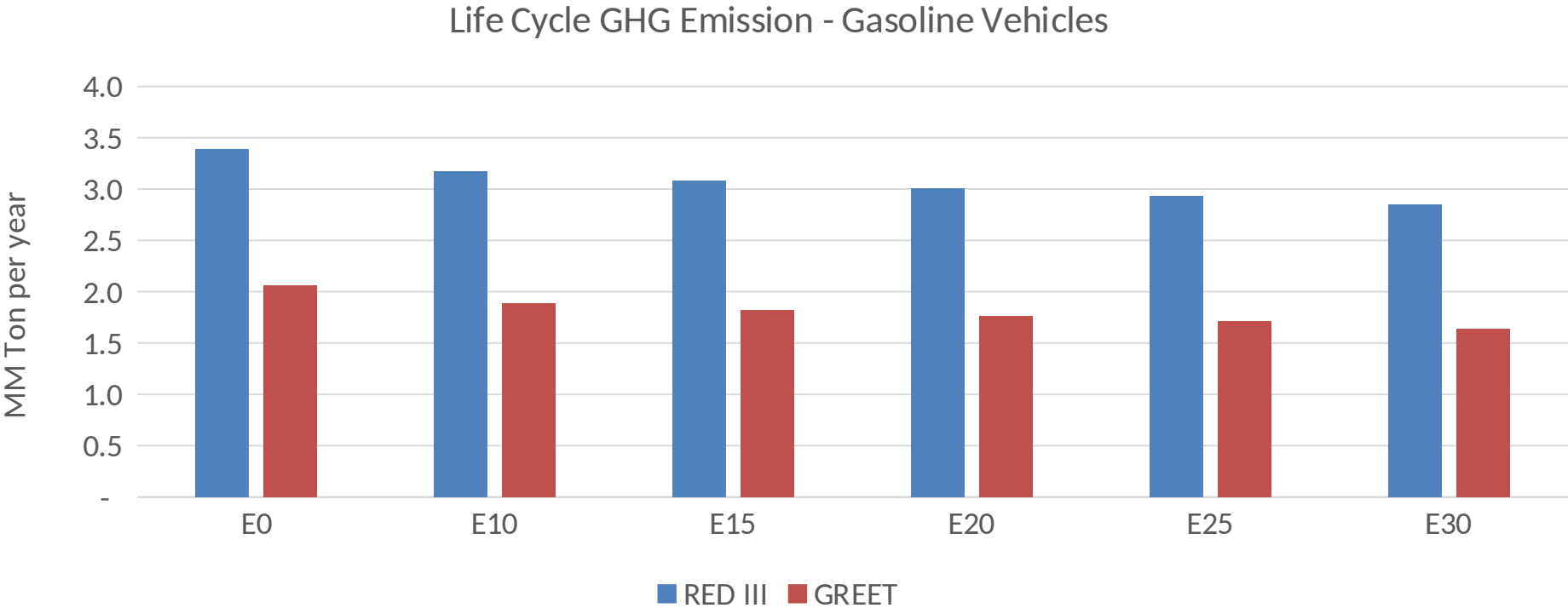
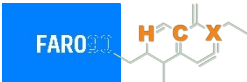
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Luxembourg – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	9.0
Target @ 2035 (MMT/year)	5.1
Delta	3.9

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.4%	11.6%	14.5%	17.1%	20.5%
RED II/III	0.0%	6.4%	8.9%	11.2%	13.3%	15.8%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.4%	6.1%	7.6%	8.9%	10.8%
RED II/III	0.0%	5.5%	7.7%	9.7%	11.4%	13.7%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90